

Overview of simulation pipeline

Hampus Österbom

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1 INTRODUCTION

This document introduces, demonstrates, and validates a simulation pipeline designed to model heterogeneous (and homogeneous) interferometric arrays in CASA. The pipeline supports arbitrary combinations of dish sizes, per-antenna aperture efficiencies, and per-baseline-group imaging, and is built on the heterogeneous-array simulation framework originally developed by R. V. Urvashi¹.

First, in Sec. 2 we will briefly discuss and lay out the core equations and relations that are the basis of this work. The key steps of the pipeline and simulations are also laid out and explained, as well as the validation pipeline plus metrics.

Secondly, we show example configurations and corresponding simulations. We start small in Sec. 3, with an array consisting of only four antennas (2×12 m and 2×7 m). We begin with this minimal four-antenna array for three reasons: (1) it produces only three distinct baseline types (12–12, 7–7, and 12–7), allowing the noise–scaling relations to be tested in complete isolation; (2) the small number of visibilities makes it easy to verify, by hand and numerically, that the simulated visibility noise, weights, and image-domain RMS follow the analytic SEFD-based predictions; and (3) it provides a controlled setting in which the effects of heterogeneous antenna diameters and efficiencies can be clearly demonstrated before moving to larger, realistic ALMA configurations.

After validating the noise model with small arrays, we progress to larger and more realistic configurations, including systems whose primary beams are not natively known to CASA. This enables us to test the creation, registration, and imaging behavior of custom voltage patterns (VPs) and their corresponding primary beams (PBs). The simulations span a range of heterogeneous arrays, dish sizes, elevations, weather conditions, and aperture efficiencies, demonstrating the robustness and flexibility of the pipeline.

2 THERMAL NOISE IN A HETEROGENEOUS INTERFEROMETER

In an heterogeneous array, the antennas have different diameters and therefore different sensitivities. This is usually expressed via the system-equivalent flux density (SEFD) of each antenna, and the thermal noise on a baseline i – j scales roughly as²

$$\sigma_{ij} = \frac{1}{\eta_s} \frac{\sqrt{\text{SEFD}_i \text{SEFD}_j}}{\sqrt{2 \Delta \nu t_{\text{int}}}}, \quad (1)$$

where η_s is the system efficiency (typically $\eta_s \simeq 0.88$ in ALMA/CASA (Cortes et al., 2025)). For a homogeneous baseline D – D , this reduces to

$$\text{SEFD}(D) = \eta_s \sigma_{DD} \sqrt{2 \Delta \nu t_{\text{int}}}, \quad (2)$$

Larger dishes (lower SEFD) give smaller σ_{ij} ; baselines involving smaller dishes are noisier.

In our pipeline, we do not manually compute σ_{ij} for each baseline. Instead, we let CASA’s noise machinery generate the vis-domain noise per antenna (via atmospheric parameters and the ATM model), which implicitly produces the correct baseline-dependent σ_{ij} for heterogeneous arrays.

2.1 How we add noise: noise first, then skymodel

The heterogeneous simulation follows this order:

1. Create an empty measurement set with the desired array configuration, pointing pattern, and spectral setup (`makeMSFrame`).
2. Attach per-antenna primary beams in a VP table (`build_per_antenna_pb_vptable`), so that later imaging and prediction know about the different dish sizes.
3. Add thermal noise in the visibility domain before any sky signal is introduced:

* If `noise_mode='atm'`, we call `addNoiseSim(msname=..., pwv_mm=..., t_ground_K=..., ...)`, which uses an atmospheric / system model to populate DATA with complex Gaussian noise.

After this step, DATA contains noise-only visibilities with the correct heterogeneity.

Importantly, because the visibilities are still noise-only at this stage, the noise can be rescaled to represent *any*

¹https://github.com/urvashirau/Simulation-in-CASA/tree/master/Heterogeneous_Array_Simulation_and_Imaging

²See <https://science.nrao.edu/facilities/vlba/docs/manuals/oss/bsln-sens>

desired aperture efficiency pattern. The helper routine `_apply_efficiency_scaling_tsysstatm` adjusts each baseline (i, j) so that its RMS scales as

$$\sigma_{ij} \propto \frac{1}{\sqrt{\eta_i \eta_j}},$$

where η_i denotes the assigned aperture efficiency of antenna i . Since the scaling is applied prior to inserting the sky model, the signal remains intact and only the thermal noise is modified. This makes it possible to prescribe physically plausible efficiency variations (e.g., reduced efficiencies for larger dishes) and thereby simulate heterogeneous arrays more realistically.

4. Optionally, we estimate effective SEFDs per dish diameter from the noisy visibilities

(`estimate_sefd_by_diameter`), as a sanity check that the injected noise scales as expected with dish size.

5. We then predict the sky model into the `MODEL_DATA` column:

- Import the input FITS image into CASA (`importfits`).
- Regrid / reshape it to match the MS (`make_startmodel_match`).
- Use `predictImager` with `gridder='mosaic'` and the per-antenna VP table to compute model visibilities in the `MODEL_DATA`.

6. Finally, we add the sky model to the noisy visibilities:

```
tb.open(ms_out, nomodify=False)

data = tb.getcol("DATA")           # noise-only
model = tb.getcol("MODEL_DATA")    # noiseless model visibilities

data += model                      # DATA <- noise + signal

tb.putcol("DATA", data)
tb.close()
```

Because the noise was generated (and, if desired, efficiency-scaled) *before* adding the sky model, the final `DATA` column represents a realistic observation: the true model visibilities plus thermal noise whose variance correctly reflects the heterogeneous per-antenna sensitivities.

2.2 Expected image-domain noise for baseline groups

For noise analysis we split the array into baseline groups:

- **A – A** : baselines between the larger dishes,
- **B – B** : baselines between the smaller dishes,
- **A – B** (or 'cross') : cross baselines.

For each group g , the pipeline computes the per-visibility noise directly from the Measurement Set using the routine `load_sigma_vis_and_N()`, which extracts the unflagged real and imaginary parts of the `DATA` column and evaluates

$$\sigma_{\text{vis},g} = \frac{1}{2} [\text{std}(\Re V) + \text{std}(\Im V)], \quad (3)$$

and denote by N_g the number of unflagged complex visibility samples.

Image noise for a single baseline group. Under natural weighting and in the absence of signal, a dirty image pixel is a weighted average of N_g independent visibility samples, each with variance $\sigma_{\text{vis},g}^2$. For equal weights within a group, the variance of the resulting image plane is therefore

$$\sigma_{\text{im},g} \simeq \frac{\sigma_{\text{vis},g}}{\sqrt{N_g}}, \quad (4)$$

which is the same expression used internally in both `compute_theoretical_rms()` and `hetero_checkVals()` when predicting group-wise image noise. This is a direct consequence of the radiometer equation and the fact that natural weighting performs an inverse-variance average of the visibilities.

Combined image noise for a heterogeneous array. An image made from all baselines jointly (A–A, B–B, and A–B) can be viewed as the optimal inverse-variance combination of the individual group images. Since the image obtained from group g has variance $\sigma_{\text{im},g}^2$, the combined variance is

$$\frac{1}{\sigma_{\text{im},\text{all}}^2} = \sum_{g \in \{A,B,\text{cross}\}} \frac{1}{\sigma_{\text{im},g}^2} = \sum_g \frac{N_g}{\sigma_{\text{vis},g}^2}, \quad (5)$$

where the second equality follows from Eq. (4). This expression is implemented directly in `compute_theoretical_rms()`, which accumulates the terms $N_g/\sigma_{\text{vis},g}^2$ for all baseline groups and evaluates $\sigma_{\text{im},\text{all}} = W_{\text{tot}}^{-1/2}$ with $W_{\text{tot}} = \sum_g N_g/\sigma_{\text{vis},g}^2$.

where the second equality follows from Eq. (4). Thus the expected rms noise of the full heterogeneous imaging run is

$$\sigma_{\text{im},\text{all}} = \left(\frac{N_A}{\sigma_{\text{vis},A}^2} + \frac{N_B}{\sigma_{\text{vis},B}^2} + \frac{N_{\text{cross}}}{\sigma_{\text{vis},\text{cross}}^2} \right)^{-\frac{1}{2}}. \quad (6)$$

Equation (4) describes how visibility-domain noise propagates into naturally weighted images, while Equation (5) is the inverse-variance combination used within `compute_theoretical_rms()` and subsequently validated by `run_noise_validation_single_field()` and `hetero_checkVals()`. These relations ensure full consistency between the analytic expectations and the numerical results obtained from the simulated Measurement Sets.

2.3 Noise validation: from visibilities to images

To check that both the noise injection and the theoretical formulas are correct, we run a noise dedicated noise-validation pipeline:

1. Create a noise-only copy of the MS:

- Copy the simulated MS to `base_noise_ms`.
- Subtract the model visibilities from the data:
 $\text{DATA} = \text{DATA} - \text{MODEL_DATA},$
so DATA now contains only the injected thermal noise.

2. Form baseline-group sub-MSs (A, B, Cross) from this noise-only MS using `mstransform`.

3. Compute theoretical image RMS:

- For each group and for the full MS, measure $\sigma_{\text{vis},g}$, and N_g from DATA.
- Compute Eq.4, and Eq.5.

4. Measure image RMS from CASA:

- For each MS (ALL, A, B, Cross) run `tclean` with:.
- `weighting='natural'`,
- `gridder='mosaic'`,
- `niter=0`,
- the same VP table as in the simulation,

This produces pure-noise residual images. We can then use `get_rms_casa` to measure the image RMS. Note that when the FoV is much smaller than the PB, as when observing at high resolution, some PB-aliasing effects will occur and can manifest at the edge of the image/FoV. Therefore, if requiring an $\text{FoV} \gg \text{PB}$, try to image at a slightly larger FoV than what is desired, and then mask out 10 – 20% of the image edges. This seems to get rid of all/most of the aliasing effects.

2.4 Definition of Validation Metrics

All validation tables in this work use a common set of visibility- and image-domain diagnostics. For each baseline group g , we compute:

- **VisStd_{sim}**: the measured per-visibility RMS from the simulated DATA column, using $\sigma_{\text{vis}} = \frac{1}{2}[\text{std}(\Re V) + \text{std}(\Im V)]$.
- **VisStd_{norm}**: the visibility RMS normalized to the reference baseline type, highlighting the expected noise scaling between dish sizes³.
- **VisStd_{calc}**: the analytic prediction based on SEFD scaling, $\sigma_{ij} \propto (\eta_i \eta_j)^{-1/2}$, or CASA’s atmospheric model when applicable.
- **#DataPts**: the number of unflagged visibilities in group g , which sets N_g in the radiometric relation $\sigma_{\text{im},g} \simeq \sigma_{\text{vis},g} / \sqrt{N_g}$.
- **Weights** ($w_{\text{calc}}, w_{\text{sim}}$): analytic and MS-reported natural weights, which should satisfy $w \propto 1/\sigma_{\text{vis}}^2$.
- σ_{calc} : the predicted image-plane noise, combining baseline groups through the inverse-variance relation $\sigma_{\text{im},\text{all}}^{-2} = \sum_g N_g / \sigma_{\text{vis},g}^2$.
- σ_{meas} : the empirically measured RMS from the pure-noise dirty image.
- **meas/calc**: the ratio $\sigma_{\text{meas}}/\sigma_{\text{calc}}$, quantifying the agreement between simulation and analytic expectation. Values near unity indicate excellent consistency.

3 IS MY IMPLEMENTATION OK?

Before we start, note that all of the following simulations used a central frequency of 343.5 GHz (with a 7.5 GHz bandwidth), a declination of -23° (unless explicitly stated otherwise) centered around transit, pww = 1.262 mm (unless explicitly stated otherwise), and a integration time of 10 s.

3.1 demo1 – $2 \times 12 \text{ m} + 2 \times 7 \text{ m}$

Fig. 1 shows the first array which will be subject to the validation tests. This minimal heterogeneous array, consisting of $2 \times 12 \text{ m}$ and $2 \times 7 \text{ m}$ antennas, produces exactly three distinct baseline classes (12–12, 7–7, and 12–7). Because each class has a different effective sensitivity, this configuration provides the simplest possible setting in which to test the predicted visibility-noise scaling, natural-weight ratios, and resulting image-domain RMS. Its small size also makes it straightforward to trace every baseline group through the full simulation and validation pipeline. The simulations can be seen in Fig. 2, and more detailed results can be seen in Table. 1.

Notice that for the noise validation when using Briggs with robust = 0.5, the 12m–12m measured RMS aligns well with the values one obtains from the ALMA Sensitivity Calculator (using the same observational parameters in the ASC). However, for the 7m–7m baselines, the noise model seems to underestimate the noise (see Table. 2). We then remember that we’ve used $\eta_{12} = \eta_7 = 0.63$ for these results. The pipeline is flexible and allows for custom per-baseline-group efficiencies, rather than assuming the same efficiency for all antennas. If we modify the antenna efficiencies and set $\eta_{12} = 0.63$, $\eta_7 = 0.66$ (correct efficiencies for 12m- and 7m at 343.5 GHz, see Cortes et al. (2025), p. 147), and run simulations for 1 h, 2 h, 4 h, and 8 h, we get for the 7m–7m group (Table. 3):

Another advantage of this noise implementation plus scaling, is that one can freely adjust the pww of the simulation. All simulations above have so far been using a pww of 1.262 mm (4th octile), but we can modify it to whatever octile we desire, and all the values should still behave as expected at that pww. We can verify this for the 7m–7m, and 12m–12m groups in demo1, by comparing our measured values for different pwws, with the expected RMS from the ASC for each group at said pww. This noise-by-weather scaling should also then

³Note that agreement between VisStd_{sim}, VisStd_{calc}, and the corresponding natural weights is expected *within each baseline type*, but not necessarily *across* baseline types. When all antennas have the same aperture efficiency (e.g. a homogeneous 12 m array), all baselines share the same thermal noise level and thus VisStd_{sim} $\approx$ VisStd_{calc} and $w_{\text{sim}} \approx w_{\text{calc}}$ for every group. For heterogeneous arrays (e.g. 12 m+7 m or 25 m+12 m), different antenna pairs have intrinsically different SEFDs, so visibility noise scales as $\sigma_{ij} \propto (\eta_i \eta_j)^{-1/2}$. In this case the baseline groups are expected to exhibit distinct values of VisStd and of the natural weights, consistent with the analytic SEFD-based predictions.

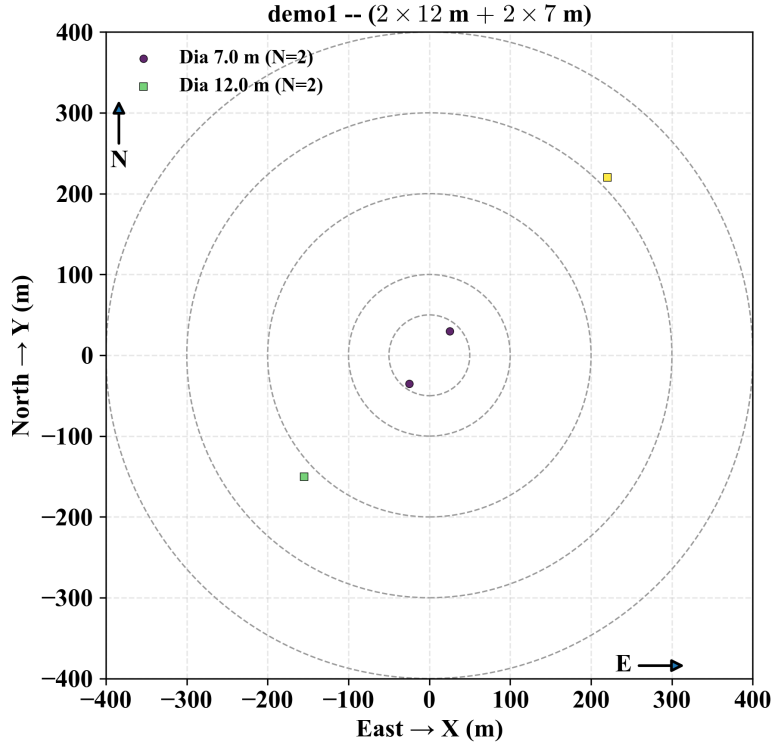


Figure 1: configuration *demo1* – consisting of $2 \times 12 \text{ m} + 2 \times 7 \text{ m}$ antennas.

Table 1: Validation metrics for baseline subsets in the heterogeneous 12m+7m configuration (natural weighting). The final column shows the ratio between measured and analytically predicted image noise. The antenna efficiencies were $\eta_{12} = \eta_7 = 0.63$.

Baseline	VisMean	Visibility Statistics			Points	Weights		Image RMS (mJy)		Ratio meas/calc
		VisStd				w		σ		
		sim	norm	calc		calc	sim	calc	meas	
(1 h): 12m–12m	0.0250	0.0136	1.000	1.0000	720	1.0000	1.0000	0.7499	0.8163	1.089
(1 h): 7m–7m	0.0744	0.0387	2.8456	2.9388	720	0.1158	0.1158	2.2081	2.4316	1.101
(1 h): 12m–7m	0.0436	0.0224	1.6471	1.7143	2880	0.3403	0.3403	0.6456	0.6977	1.081
(1 h): All	—	—	—	—	—	—	—	0.4777	0.5167	1.082
(8 h): 12m–12m	0.0280	0.0153	1.000	1.0000	5760	1.0000	1.0000	0.2978	0.2872	0.964
(8 h): 7m–7m	0.0844	0.0454	2.9673	2.9388	5760	0.1158	0.1158	0.8925	0.9107	1.020
(8 h): 12m–7m	0.0486	0.0263	1.7190	1.7143	23040	0.3403	0.3403	0.2577	0.2362	0.917
(8 h): All	—	—	—	—	—	—	—	0.1904	0.1760	0.924

Table 2: Validation metrics for a test configuration using two 12 m and two 7 m antennas, imaged with Briggs weighting ($r_b = 0.5$) and antenna efficiencies $\eta_{12} = \eta_7 = 0.63$.

Baseline	Visibility Statistics				Points	Weights		Image RMS (mJy)		Ratio meas/ASC
	VisMean	VisStd		w		σ				
		sim	norm			calc	ASC	meas		
(1 h): 12m–12m	0.0250	0.0136	1.000	1.0000	720	1.0000	1.0000	0.850	0.8602	1.012
(1 h): 7m–7m	0.0744	0.0387	2.825	2.9388	720	0.1158	0.1158	2.390	2.6971	1.128
(1 h): 12m–7m	0.0436	0.0224	1.635	1.7143	2880	0.3403	0.3403	—	0.7533	—
(1 h): All	—	—	—	—	—	—	—	—	0.5862	—
(8 h): 12m–12m	0.0280	0.0153	1.000	1.0000	5760	1.0000	1.0000	0.301	0.298	0.990
(8 h): 7m–7m	0.0844	0.0454	2.955	2.9388	5760	0.1158	0.1158	0.845	0.981	1.161
(8 h): 12m–7m	0.0486	0.0263	1.7143	1.7143	23040	0.3403	0.3403	—	0.266	—
(8 h): All	—	—	—	—	—	—	—	—	0.2005	—

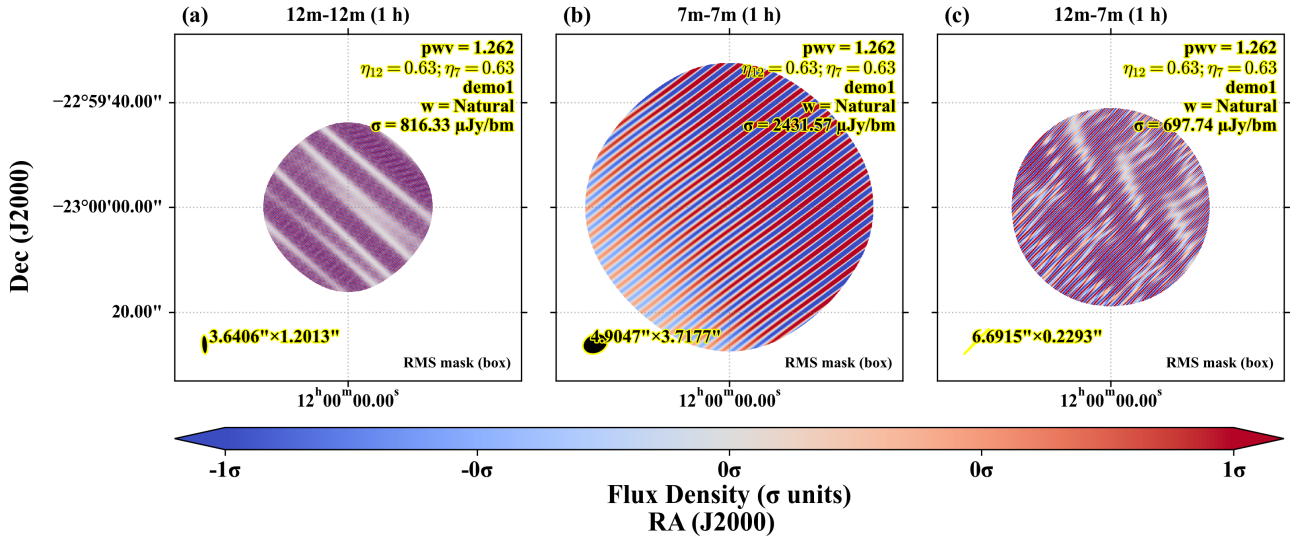


Figure 2: Noise-only dirty images for the three baseline subsets in demo1 (12–12, 7–7, 12–7) under natural weighting. Each panel is plotted in units of the measured σ for that group. The differing fringe patterns reflect the unique uv -coverage and noise levels of each baseline type.

Table 3: Expected (σ_{ASC}) and measured (σ_{meas}) image RMS for the 7 m–7 m baseline group in configuration demo1, at different on–source times. Now the efficiency for the 7 m antennas are $\eta_7 = 0.66$. All RMS values are in mJy beam^{-1} . Briggs weighting with robust = 0.5 was used in order to be able to directly compare to the ASC.

Baseline	Quantity	1 h	2 h	4 h	8 h
7m–7m	σ_{ASC}	2.39	1.69	1.2	0.85
	σ_{meas}	2.57	1.88	1.14	0.94
	$\sigma_{\text{meas}}/\sigma_{\text{ASC}}$	1.08	1.11	0.95	1.11

Note that the 7 m–7 m subset contains only a single baseline, so the noise estimate suffers from relatively large statistical fluctuations and a less stable dirty beam. As the integration time increases, the overall trend of σ_{meas} follows the expected $\sigma_{\text{ASC}} \propto t^{-1/2}$ scaling, but deviations of order $\sim 5\text{--}10\%$ at any given time are expected from finite sampling and the limited number of independent beams in the image (single pointing is being used).

be independent of the dish sizes used, but by using 7m- and 12m, we can directly compare it to the ASC (see Table. 4).

Table 4: Comparison of expected and measured RMS for two baseline groups (1 h integration) at PWV = 2.748 mm and 5.186 mm. All RMS values in mJy beam^{-1} . Briggs weighting with robust = 0.5 was used and efficiencies were $\eta_{12} = 0.63$, $\eta_7 = 0.66$.

Baseline Group	PWV = 2.748 mm		PWV = 5.186 mm	
	σ_{ASC}	σ_{meas}	σ_{ASC}	σ_{meas}
12m–12m	1.40	1.46	2.71	2.84
7m–7m	3.92	4.36	7.61	8.50

3.2 *demo2* – $10 \times 12 \text{ m} + 4 \times 25 \text{ m}$

Now we define the configuration *demo2* (Fig. 3), which consists of $10 \times 12 \text{ m} + 4 \times 25 \text{ m}$ antennas. This configuration extends the analysis to a larger heterogeneous configuration with dishes whose primary beam is not natively known to CASA (25 m). The resulting simulations can be seen in Fig. 4, and a more detailed look at the results can be found in Table. 5.

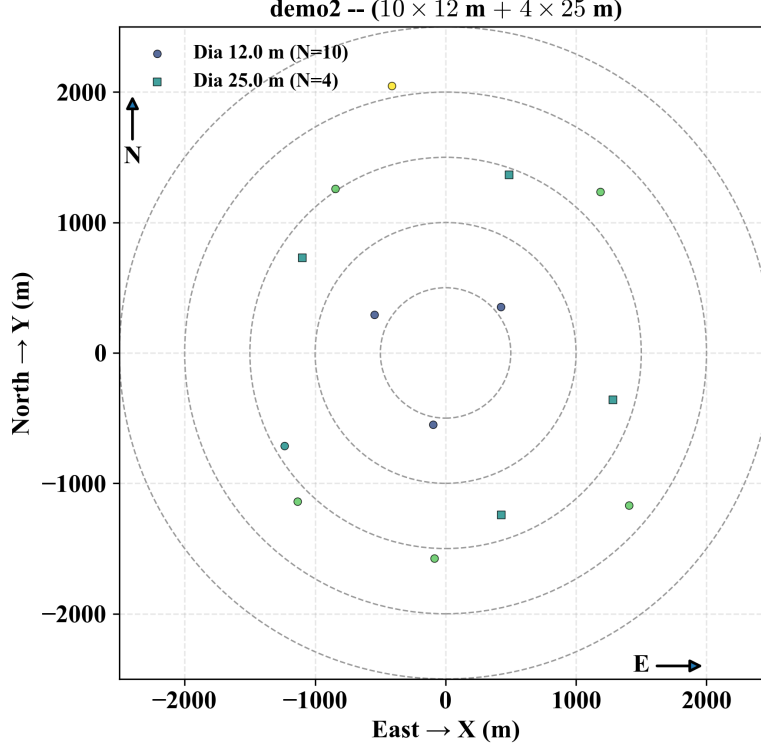


Figure 3: Configuration *demo2*, which consists of $10 \times 12 \text{ m} + 4 \times 25 \text{ m}$ antennas.

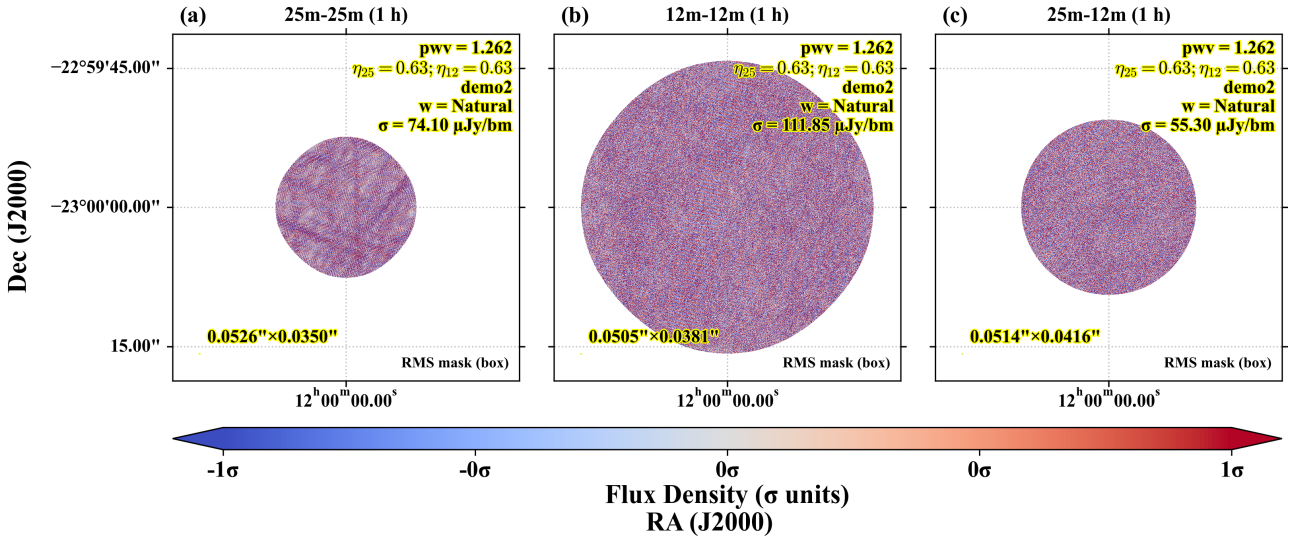


Figure 4: Noise-only dirty images for *demo2* baseline groups (25–25, 12–12, 25–12) at 1 h integration. The image RMS closely follows the analytic predictions shown in Table 5.

Table 5: Validation metrics for a test configuration using ten 12 m and four 25 m antennas, imaged with natural weighting and antenna efficiencies $\eta_{12} = \eta_{25} = 0.63$.

Baseline	Visibility Statistics				Points	Weights		Image RMS (mJy)		Ratio meas/calc
	VisMean	VisStd				w	σ			
		sim	norm	calc			calc	sim	calc	
(1 h): 25m–25m	0.0058	0.0030	1.000	1.0000	4320	1.0000	1.0000	0.0700	0.0741	1.059
(1 h): 12m–12m	0.0251	0.0132	4.400	4.3403	32400	0.0531	0.0531	0.1114	0.1119	1.004
(1 h): 25m–12m	0.0120	0.0063	2.100	2.0833	28800	0.2304	0.2304	0.0564	0.0553	0.980
(1 h): All	—	—	—	—	—	—	—	0.0408	0.0413	1.012

3.3 demo 3 – larger arrays

Now we go even larger – simulating a pure noise image with an extended heterogeneous array, consisting of 43×12 m (C43-10) and 6×50 m antennas, see Fig. 5. To be semi-realistic, and to show the capabilities of the efficiency scaling, we set $\eta_{12} = 0.63$, $\eta_{50} = 0.504$ at 343.5 GHz. The results can be seen in Table. 6 and the corresponding simulations in Fig. 6.

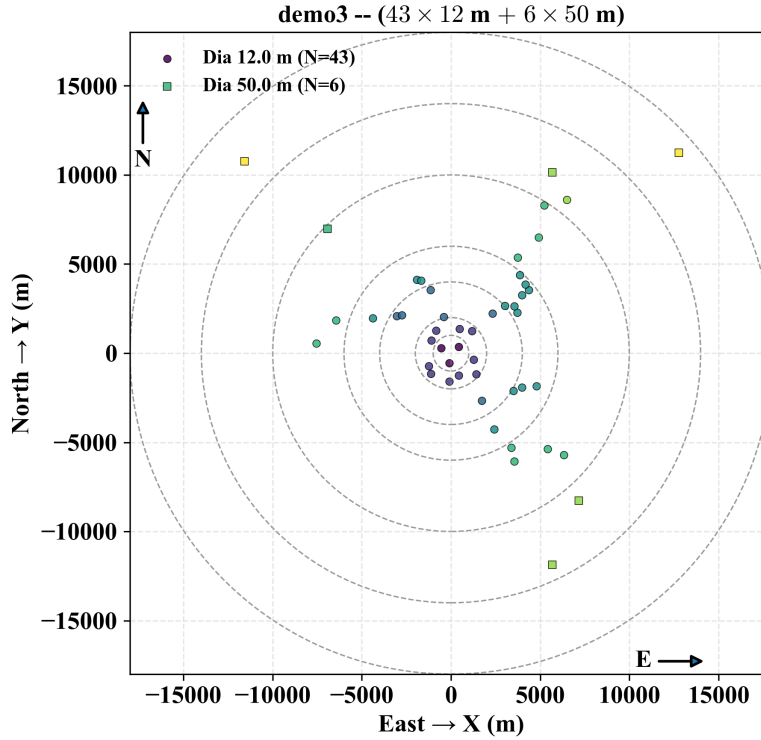


Figure 5: configuration *demo3* – consisting of 43×12 m + 6×50 m antennas.

Table 6: Validation metrics for mixed 50 m and 12 m baseline subsets (natural weighting, $\eta_{12} = 0.63$, $\eta_{50} = 0.504$). The $\text{VisStd}_{\text{sim}}$, $\text{VisStd}_{\text{calc}}$, and weight differences follow from the heterogeneous antenna sizes used in the simulation.

Baseline	Visibility Statistics				Points	Weights		Image RMS (mJy)		Ratio meas/calc
	VisMean	VisStd		w		σ				
	sim	norm	calc				calc	sim	calc	meas
(1 h): 50m–50m	0.0018	0.0009	1.000	1.0000	108000	1.0000	1.0000	0.0138	0.01377	0.998
(1 h): 12m–12m	0.0251	0.0131	14.5555	17.3611	650160	0.0033	0.0052	0.0248	0.0240	0.968
(1 h): 50m–12m	0.0067	0.0035	3.8888	4.1667	185760	0.0576	0.0720	0.0124	0.0120	0.968
(1 h): All	—	—	—	—	—	—	—	0.0087	0.0088	1.01

To show that the weighting (and Std) deviations truly come from the fact that we changed the antenna efficiency for the 50 m dishes, we can set $\eta_{12} = \eta_{50} = 0.63$ and re-run the simulations. The simulations can be seen in Fig. 7, and the results in Table. 7.

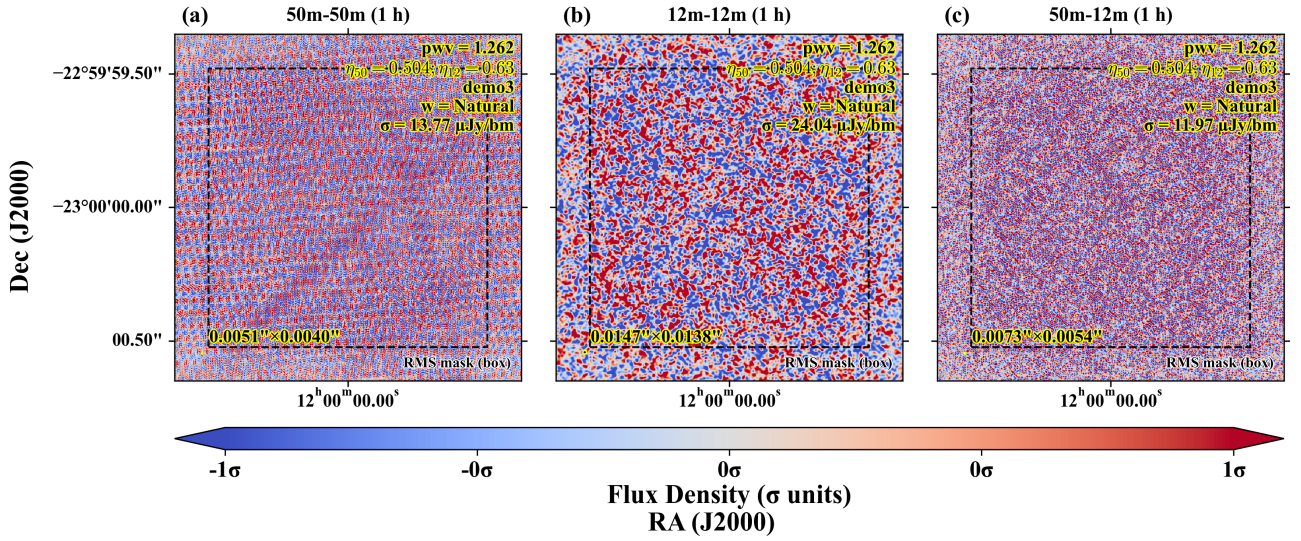


Figure 6: Noise-only dirty images for the large heterogeneous array demo3 (43×12 m + 6×50 m).

Table 7: Validation metrics for mixed 50 m and 12 m baseline subsets (natural weighting, $\eta_{12} = \eta_{50} = 0.63$).

Baseline	VisMean	Visibility Statistics			Points	Weights		Image RMS (mJy)		Ratio meas/calc
		sim	norm	calc		calc	sim	calc	meas	
(1 h): 50m–50m	0.0014	0.0007	1.0000	1.0000	108000	1.0000	1.0000	0.0110	0.01102	1.002
(1 h): 12m–12m	0.0251	0.0131	18.7143	17.3611	650160	0.0033	0.0033	0.0248	0.0240	0.968
(1 h): 50m–12m	0.0060	0.0031	4.091	4.1667	185760	0.0576	0.0576	0.0111	0.0107	0.964
(1 h): All	—	—	—	—	—	—	—	0.0075	0.0071	0.947

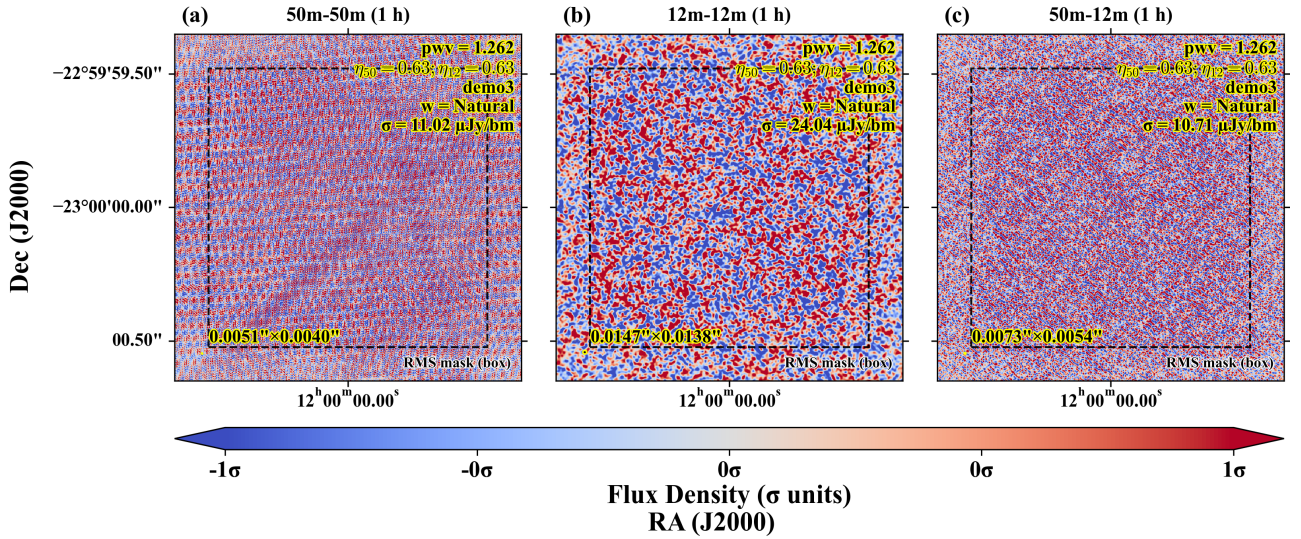


Figure 7: Same as Figure 6 but with equalized efficiencies ($\eta_{12} = \eta_{50} = 0.63$), demonstrating that the pipeline correctly adjusts baseline noise when aperture efficiencies are changed.

For even more example simulations and results with varying configurations, see Sec. 5.1.

4 TECHNICAL OVERVIEW

This section of the document describes the full ALMA simulation pipeline implemented across the modules `run_simulation_pipeline.py`, `qol_pipeline.py`, `pb_vp.py`, `generate_builtin_sky_model.py`, `utils.py`, and `validation.py`. The pipeline supports both homogeneous and heterogeneous ALMA arrays, full per-antenna voltage-pattern (VP) modelling, atmospheric noise simulation, CLEAN/dirty imaging, baseline-group imaging, and noise-validation. The goal is to provide a clear, concise technical summary of how the system works and how it should be used.

Pipeline Architecture

The pipeline contains five major subsystems, each implemented primarily in one or more Python modules:

1. **Main** (`run_simulation_pipeline.py`): top-level driver that reads the user / example settings, resolves ALMA configuration files, computes imaging geometry, generates or imports the sky model, decides between heterogeneous vs homogeneous simulation paths, calls the appropriate MS-creation + imaging + validation routines, and finally writes a structured `pipeline_log.json` summary for the run.
2. **Simulation & Imaging Core** (`qol_pipeline.py`): low-level CASA utilities that actually build custom MSs, construct hour-angle scan patterns, apply the `tsys-atm` noise model with per-diameter efficiency scaling, predict the sky model into `MODEL_DATA`, add it to the noisy `DATA`, estimate SEFDs from the visibilities, compute array-dependent imaging geometry (beam, cell, imsize, hetero vs homo), and provide robust image-statistics tools (PB-aware RMS, peak, restoring beam, baseline grouping, etc.).
3. **Built-in Sky Model Generator** (`generate_builtin_sky_model.py`): standalone CASA script used by the pipeline when `sky_mode="builtin"`. It generates two synthetic test sources on the requested grid: (i) a “ring” model with a bright central Gaussian and eight ring Gaussians that are each 100x fainter than the centre, and (ii) a “point+disk” model with a bright central point source plus a fainter compact disk. The script can be driven either from JSON configuration or from command-line arguments.
4. **PB/VP System** (`pb_vp.py`): tools for primary-beam and voltage-pattern handling. This module computes λ/D -based PB sizes, generates per-dish PB images (ALMA Gaussian, Airy, or custom Gaussian) on a common grid with guard rings, groups antennas by diameter, and assembles a per-antenna VP table that `tclean` can use for AW-projection in both heterogeneous and “PB-aware” homogeneous imaging.
5. **Imaging/Stats Helpers & General Utils** (`utils.py` and `qol_pipeline.py` and): convenience layer around CASA imaging. This includes FFT-friendly imsize selection, dirty and CLEAN imaging wrappers, automatic multiscale setup from the PSF, central-box mask generation, robust PB-masked RMS estimators, baseline-group imaging (A/B/cross), as well as generic helpers for logging, safe removal of CASA products, and short, descriptive image name tags.
6. **Noise Validation & Heterogeneity Checks** (`validation.py`): validation suite that copies the science MS into a noise-only sandbox, subtracts the sky model (or plain `simobserve` MS) from the `DATA` column, constructs A/B/cross sub-MSs where relevant, computes theoretical σ_{vis} and σ_{im} for each baseline group, runs natural-weight mosaic dirty imaging, measures PB-masked image RMS with the same machinery as the main pipeline, and prints/tabulates measured vs theoretical noise plus per-baseline-type diagnostics for heterogeneous arrays.

End-to-End Workflow

The complete pipeline executes the following sequence:

1. Load user settings and global configuration.
2. Locate ALMA configuration file and parse antenna positions & dish diameters.
3. Automatically compute imaging geometry: beam FWHM, cell size, FFT-efficient imsize.

4. Generate sky model via CASA script and import ⁴.
5. Create Measurement Set (MS): heterogeneous (custom) or homogeneous (simobserve).
6. Generate per-diameter PB images and construct per-antenna VP table.
7. Apply atmospheric noise (tsys-atm) and per-diameter efficiency scaling.
8. Predict sky model (MODEL_DATA) and add to noisy visibilities.
9. Perform dirty imaging, CLEAN if requested.
10. Optionally image baseline groups (A/A, B/B, A/B).
11. Run noise-validation imaging for heterogeneous configurations.
12. Export FITS & log results.

4.1 *Heterogeneous vs Homogeneous Arrays*

Dish diameters parsed from ALMA `.cfg` files determine the simulation mode.

4.1.1 Homogeneous Arrays

- One single dish size.
- MS generated with `simobserve`.
- Optional VP table for PB-aware imaging (only ever needed when simulating homogeneous arrays with other dish sizes than 7 m and 12 m!).
- Uniform efficiency noise scaling.

4.1.2 Heterogeneous Arrays

- Multiple dish sizes present (e.g., 12m+25m).
- Custom MS preserving per-antenna diameters.
- PB/VP image per diameter.
- Per-baseline noise scaling for correct SEFD ratios.
- Baseline-group imaging & validation.

4.2 *Primary-Beam & Voltage-Pattern System*

The PB/VP system generates accurate per-antenna PB models:

1. Compute PB FWHM via ALMA or analytic models.
2. Generate PB images (voltage form, XX/YY)⁵.
3. Construct VP table with `vpmanager.setpbimage`.
4. Enable AW-projection in `tclean` for heterogeneous imaging.

⁴It is possible to supply your own skymodel/source as a fits file, as when normally simulating with CASA. A "skyscript" is just a script which crates a skymodel/source and base the dimensions of it upon the input parameters in the pipeline, it can therefore dynamically adjust the dimensions based on the configuration used.

⁵**Why XX/YY?** CASA's voltage-pattern tables expect per-polarization primary beams. For ALMA, only the parallel-hand feeds (XX, YY) are used for PB correction; cross-hands (XY, YX) are set to zero.

4.3 Built-in Sky Models

The pipeline includes two synthetic sky models implemented in `generate_builtin_sky_model.py`. These models provide controlled, reproducible test cases without requiring external FITS files.

- **9-Gaussian Ring Model** This model consists of nine concentric Gaussian components: a central compact Gaussian that is 100 times brighter than the surrounding rings, plus eight progressively larger, fainter Gaussians arranged radially. The result is a bright compact core embedded in smooth extended emission, ideal for testing dynamic-range limits, multiscale CLEAN behaviour, and PB-correction performance.
- **Point+Disk Model** A single unresolved point source at the image center combined with a circular Gaussian disk of adjustable size and flux ratio. This model is useful for validating flux recovery, beam characterization, and deconvolution behaviour on mixed compact–extended emission.

Both models are generated as single-channel CASA images using the pipeline’s imaging geometry (cell size, imsize, reference frequency), ensuring that their angular scales and flux distributions remain consistent with the user’s configuration.

4.4 Noise Injection

Noise injection is a two-step process:

1. Atmospheric noise (tsys-atm) using reference efficiency.
2. Per-diameter scaling: for baseline $i-j$,

$$f_{ij} = \frac{\eta_{\text{ref}}}{\sqrt{\eta_i \eta_j}}.$$

Extra Gaussian noise added if $f > 1$, DATA scaled if $f < 1$.

4.5 Imaging

1. Dirty imaging using mosaic gridding & AW-projection⁶.
2. Compute RMS & CLEAN threshold.
3. Optional CLEAN with multiscale (with dynamically scales based upon the beam size)⁷ and custom mask.
4. Measure peak, RMS, beam size.

4.5.1 Baseline-Group Imaging

For heterogeneous arrays, A, B, and cross baselines are imaged separately (if turned on). Dirty and optional CLEAN images are produced and validated.

4.6 Noise Validation

1. Copy MS and subtract model → noise-only MS.
2. Build A, B, cross sub-MSs.
3. Compute theoretical σ_{im} per group.
4. Combine groups via inverse-variance weights.
5. Perform natural-weight dirty imaging and measure RMS.
6. Compare with theoretical predictions.

⁶Mosaic gridding and AW-projection is needed for accurate heterogeneous array simulations independently from the number of pointings used! It enables PB aware imaging, which will use the custom PBs and VPs one created.

⁷Multiscale with dynamically set scales is recommended; although, the pipeline allows for any deconvolver to be used!

4.7 Running the Pipeline

- (i) Make sure the following files (and any others the pipeline imports) are in the same directory.
- (ii) Set the CASA binary path, e.g., `casa_bin = "/path/to/your/casa/bin/casa"`
- (iii) Adjust parameters in `EXAMPLE_SETTINGS`.
- (iv) Run the pipeline using CASA in non-GUI mode:

```
casa --nogui --nologger -c run_simulation_pipeline.py
```

All MSs, FITS, VP tables, logs, and JSON summaries are saved in `project_base`.

4.7.1 Example Configuration: `EXAMPLE_SETTINGS`

The publicly available main script is currently controlled through a single Python dictionary, `EXAMPLE_SETTINGS` (found at the top of `run_simulation_pipeline.py`), which groups all user-facing parameters in one place:

```
EXAMPLE_SETTINGS = dict(

    # --- Paths / project structure ---
    project_base="ALMA.validate",          # Folder where all outputs go
    casa_bin="/home/hampus/casa-6.7.0-31-py3.10.el8/bin/casa", # CASA executable
    extra_dirs=[],                          # Extra dirs to search for cfg files

    # --- Array configuration(s) ---
    config_map={"alma.H200.cfg": "GG"},     # {cfg_file : short_tag}
    declinations=["-23d00m00.00"],          # Target declination(s)
    integration_times=["1h"],               # On-source time(s)

    # --- Frequency setup ---
    center_freq="343.5GHz",                 # Reference frequency
    width="7.5GHz",                         # Total bandwidth
    pwv=1.262,                             # PWV for ATM noise

    # --- Imaging / sky setup ---
    phasecenter_ra="12h00m00.00s",          # Phase center RA
    fov_arcsec=1.28,                       # Desired field of view (arcsec)
    sampling_factor=5.0,                   # Beam/cell sampling (higher -> smaller pixels)

    # --- Sky model selection ---
    # sky_mode controls *where* the sky model comes from:
    # "builtin" : use one of the internal test skies (sky_type below)
    # "fits"    : use a user-supplied FITS image directly
    sky_mode="builtin",

    # If sky_mode == "builtin", choose between:
    # "ring"      : 8+1 Gaussian ring (central Gaussian + 8 around, ring Gaussians are 100x fainter)
    # "pointdisk" : bright central point source + faint small disk
    sky_type="ring",

    # If sky_mode == "fits", this must point to an existing FITS file
    # that will be imported and used as the sky brightness model.
    sky_fits=None,                          # e.g. "/path/to/my_sky_model.fits"

    # --- Built-in sky parameters (only used for sky_mode="builtin") ---
    # Ring model (sky_type="ring"):
    ring_flux=30e-3,                         # Jy of central Gaussian (set VERY low for an
    essentially noise-only image!)           # Number of Gaussians in the ring
    ring_n_sources=8,                        # Ring radius in arcsec
    ring_radius_as=0.15,                     # Major axis of each Gaussian (arcsec)
    ring_major_as=0.035,                    # Minor axis of each Gaussian (arcsec)
    ring_minor_as=0.035,                    # Position angle of Gaussians (deg)
    ring_pa_deg=0.0,

    # Point+disk model (sky_type="pointdisk"):
    pnd_flux=1.0,                           # Base total flux (Jy) (set VERY low for an
    essentially noise-only image!)           # If None -> uses pnd_flux
    pnd_point_flux=None,                     # If None -> uses 0.1 * pnd_flux
    pnd_disk_flux=None,                      # Disk diameter (arcsec),
    pnd_disk_diam_as=0.10,
```

```

# --- Optional manual geometry overrides ---
# The pipeline normally computes "good" imaging geometry automatically
# (cell size + image size) from the array beam and fov_arcsec.
# These two knobs let you override that behaviour:
#
# * If cell_arcsec is not None:
#   - this fixed cell size (arcsec / pixel) is used for BOTH
#     the sky model and all imaging.
# * If imsize_manual is not None:
#   - this fixed image size (pixels) is used for BOTH the
#     sky model and all imaging.
#
# If either is None, the auto-geometry from compute_imaging_geometry()
# is used for that parameter instead.
cell_arcsec=None,          # Fixed cell size (arcsec), overrides auto if set
imsize_manual=None,        # Fixed image size (pixels), overrides auto if set

# --- tclean / deconvolution ---
weightings=["briggs"],      # Weighting scheme(s)
robust_values=[0.5],        # Briggs robust parameter(s)
uvtaper_values=["0.0arcsec"], # UV taper(s)
niter=500,                  # CLEAN iterations (0 = dirty)
deconvolver="multiscale",   # Deconvolver type
threshold_factor=2,         # CLEAN threshold = factor x dirty RMS

# global pblimit used for all tclean calls that specify pblimit
pblimit=0.2,
# Cleaning mask settings
clean_use_mask=True,        # If True, use a central box mask for CLEAN (only
if niter>1)
clean_mask_center_frac=0.1, # Fraction of image size to keep as central mask

# --- Noise / Primary Beam ---
noise_mode="atm",           # 'atm', 'manual', or 'none' (atm is HIGHLY
recommended)

# --- VP / PB usage for homogeneous configs ---
use_vptable_in_homo=True,   # If True: build/use a VP table even for
homogeneous arrays (PB-aware imaging)

# Antenna/aperture efficiencies for tsys-atm noise model // Eff. for 7-and 12m at various freqs
# can be found in
# the ALMA technical handbook page 147 (cycle 12 handbook)
eta_A=0.63,                 # Efficiency for large dishes (e.g. 12m)
eta_B=0.66,                 # Efficiency for small dishes (e.g. 7m)
spilloverefficiency=0.96,   # Spillover efficiency
trx_K=None,                 # Receiver temperature (K) ; None => auto from
ALMA band & center_freq
correfficiency=0.88,        # Correlator efficiency
waterheight=2000.0,         # Water vapour scale height (m)

pb_model="alma",            # 'alma', 'airy', or 'gauss' ('alma' is highly
recommended, but you can try 'airy' as well)

# --- Validation / diagnostic "buttons" ---
validate_noise_theory=True,  # Run noise-validation imaging?
noiseval_field="0",          # FIELD id used for noise validation (keep as 0,
only 0 is currently supported)
noiseval_do_groups=True,    # For heterogeneous arrays: also do A/B/cross
groups?
print_vptable=False,        # If True: prints the combined vptable with the
PB mapping and which antennas use which PB.

image_baseline_groups=True,  # Also image A/B/cross groups for the main run?

# --- Logging / cleanup ---
log_file="pipeline_test.log", # Log file (None = stdout only)
clean_project_base=True,      # Remove previous outputs before run

# --- RMS measurement settings (used in measure_image_stats) ---
# PS: note that when FoV << PB, aliasing will occur, for accurate RMS, one should mask the
# edges of the image!
rms_mode="auto",            # 'full' = whole image, 'pb' = PB-limited region,
'auto' = apply masks below (PB mask + geometric masks)
rms_use_pb=True,            # If True: apply PB mask (keep pixels where PB >=
rms_pb_min)

```

```

rms_pb_min=0.05,                                # Primary-beam cutoff for RMS regio
# --- Geometric masking controls ---
rms_center_frac=0.1,                             # Exclude a central box of this fractional size
                                                # (e.g., 0.10 -> remove inner 10% area; useful to
mask_bright_source_cores)

rms_keep_center_frac=0.8,                         # Keep only a central box of this fractional size
(crops_away_noisy_edges).

rms_center_frac are set:                         # If both rms_keep_center_frac AND
                                                # -> RMS is computed in an ANNULUS-LIKE region:
                                                # central box defined by keep_center_frac
                                                # MINUS the inner excluded box defined by
center_frac.
# Summary of combinations:
#   Only rms_center_frac set:      mask *only* the inner core.
#   Only rms_keep_center_frac set: use *only* central part of image; discard edges.
#   Both set:                      keep middle region but punch out the inner core
#                                   (annulus / donut mask).
#   Neither set:                   no geometric masking (PB mask only, if enabled).
rms_nsigma_clip=5.0,                  # Sigma clipping for robust RMS estimation
rms_max_points=2_000_000,             # Max number of pixels sampled for RMS
)

```

In practice:

- Paths, array configuration, and frequency setup control which ALMA config, declination, and bandwidth are used.
- Imaging/sky options set the FoV, pixel sampling, and which built-in source that is to be used (or your own supplied source, as a fits file).
- `tclean` parameters (weighting, robust, taper, niter) set the imaging style and deconvolution depth.
- Noise and PB settings define atmospheric conditions, aperture efficiencies, and the analytic PB model.
- Validation flags toggle noise-validation imaging and baseline-group checks.
- RMS settings control how image noise is measured (PB mask, central exclusion, sigma-clipping).

4.8 Noise implementation and aperture–efficiency scaling

Thermal noise is injected using a two–stage procedure. First, CASA’s `tsys-atm` model generates visibility-domain noise that already includes the correct dependence on dish diameter, atmospheric conditions, and receiver temperature. Second, an optional per-baseline efficiency rescaling is applied to introduce arbitrary per-antenna aperture efficiencies $\{\eta_i\}$, which `tsys-atm` cannot represent on its own (it accepts only a single global aperture efficiency parameter). The sky model is added *after* these noise operations, ensuring that only the thermal noise is modified.

4.8.1 Stage 1: Atmospheric noise from `tsys-atm`

CASA’s `tsys-atm` model computes the system temperature using an atmospheric radiative-transfer calculation and the Brown (1995) MMA sensitivity formulation.⁸ Crucially, this model incorporates the dish diameter D_i for each antenna (as stored in the MS) when evaluating the sensitivity. As a result, baselines between smaller dishes (e.g. 7 m–7 m) naturally exhibit higher per-visibility noise than baselines between larger dishes (e.g. 12 m–12 m), with mixed baselines lying in between. This behaviour is clearly visible in our validation results (Tables 1–2 and 5–7), where the measured visibility RMS and natural weights follow the expected SEFD-like scaling $\sigma_{ij} \propto D_i^{-1} D_j^{-1}$.

The `tsys-atm` task accepts only a single global aperture efficiency η_{ref} during this stage. After noise injection, the `DATA`, `SIGMA`, and `WEIGHT` columns therefore contain thermal noise that correctly reflects dish-diameter and atmospheric effects, but assumes identical efficiency for all antennas.

⁸See CASA documentation for `setnoise(mode='tsys-atm')`.

```

sm.openfromms(msname)
sm.setnoise(mode='tsys-atm', antefficiency=eta_ref, ...)
sm.corrupt()
sm.close()

```

4.8.2 Stage 2: Per-baseline efficiency scaling

Before applying the efficiency corrections, we introduce the reference efficiency η_{ref} , which is the single aperture-efficiency value supplied to `tsys-atm` during Stage 1. While `tsys-atm` correctly accounts for the actual dish diameters in the array, it accepts only one global aperture efficiency, and therefore the noise it produces corresponds to an array in which all antennas share this same efficiency. We set this reference value to $\eta_{\text{ref}} = \min(\eta_i, \eta_j)$, so that the initial CASA-generated noise is never overly optimistic: the noisiest (least efficient) antennas require no further adjustment, whereas baselines involving higher-efficiency antennas will subsequently have their noise reduced. The scaling factor in Eq. (7) thus quantifies how the true baseline sensitivity differs from the uniform-efficiency case assumed by `tsys-atm`.

To introduce heterogeneous aperture efficiencies, each antenna i is assigned its desired efficiency η_i . For each visibility row corresponding to baseline $i-j$, we define the multiplicative scaling factor

$$f_{ij} = \frac{\eta_{\text{ref}}}{\sqrt{\eta_i \eta_j}}. \quad (7)$$

Let σ_{ref} denote the existing per-visibility noise RMS as reported in `SIGMA`. The target noise is

$$\sigma_{\text{true}} = f_{ij} \sigma_{\text{ref}}. \quad (8)$$

Case $f_{ij} > 1$: adding noise. If the baseline should be *noisier* than the reference, the desired variance is⁹

$$\sigma_{\text{true}}^2 = \sigma_{\text{ref}}^2 + \sigma_{\text{extra}}^2, \quad \sigma_{\text{extra}} = \sigma_{\text{ref}} \sqrt{f_{ij}^2 - 1}. \quad (9)$$

A zero-mean circular complex Gaussian random variable $n_{ij} \sim \mathcal{CN}(0, \sigma_{\text{extra}}^2)$ is then added (Thompson et al., 1986, pp. 47, 407–408):

$$V \leftarrow V + n_{ij}. \quad (10)$$

Case $f_{ij} < 1$: reducing noise. If the baseline should be *less* noisy, we scale the visibility:

$$V \leftarrow f_{ij} V, \quad \sigma_{\text{true}} = f_{ij} \sigma_{\text{ref}}. \quad (11)$$

Since the visibilities contain noise only at this stage, this reduces the thermal noise without affecting any sky signal (which is added later).

Example (12 m, 7 m, and mixed baselines). For the `demo1` efficiencies

$$\eta_{12} = 0.63, \quad \eta_7 = 0.66, \quad \eta_{\text{ref}} = \eta_{12},$$

the baseline factors become

$$f_{12-12} = 1.00, \quad f_{7-7} \simeq 0.95, \quad f_{12-7} \simeq 0.98.$$

Thus 12 m–12 m baselines retain the original `tsys-atm` noise, 7 m–7 m baselines receive a small noise reduction, and mixed 12 m–7 m baselines fall in between. This yields the expected ordering

$$\sigma_{12-12} < \sigma_{12-7} < \sigma_{7-7},$$

consistent with the measured visibility RMS in Tables 1–2 and 5–7.

⁹The expression for σ_{extra} follows from adding independent Gaussian variances: $\sigma_{\text{true}}^2 = \sigma_{\text{ref}}^2 + \sigma_{\text{extra}}^2$, with $\sigma_{\text{true}} = f_{ij} \sigma_{\text{ref}}$.

4.8.3 Simulation ordering

1. Apply `tsys-atm` noise using a single reference efficiency η_{ref} (dish-diameter dependence included by CASA).
2. Apply the per-baseline efficiency scaling described above.
3. Predict the sky model into `MODEL_DATA` and add it to `DATA`.

Thus the efficiency scaling affects only the thermal noise, while the sky model is added afterwards with its original, unscaled amplitude.

4.8.4 SEFD estimation as a validation check

Although the pipeline does not use SEFDs explicitly during noise injection, SEFD provides a convenient physical benchmark for verifying that the simulated visibility noise scales correctly with antenna diameter and efficiency. For a baseline $i-j$, the radiometer equation gives

$$\sigma_{ij} \propto \frac{\sqrt{\text{SEFD}_i \text{SEFD}_j}}{\sqrt{2 \Delta \nu t_{\text{int}}}},$$

so that larger, more efficient dishes (lower SEFD) produce smaller visibility RMS. In a homogeneous array this reduces to $\text{SEFD}(D) = \eta_s \sigma_{DD} \sqrt{2 \Delta \nu t_{\text{int}}}$, but in a heterogeneous array each diameter produces a distinct effective SEFD.

After applying `tsys-atm` noise and the per-baseline efficiency scaling, we estimate an “effective SEFD” for each dish diameter directly from the measured visibility RMS in the noise-only MS:

$$\text{SEFD}_{\text{est}}(D) \propto \sigma_{DD} \sqrt{2 \Delta \nu t_{\text{int}}}.$$

These estimates are not used in the simulation itself, but provide a sanity check: the recovered ratios $\text{SEFD}_{\text{est},12}/\text{SEFD}_{\text{est},7}$, $\text{SEFD}_{\text{est},25}/\text{SEFD}_{\text{est},12}$, etc., should match the analytic expectation $\sigma_{ij} \propto (D_i D_j)^{-1}$ (or the prescribed efficiencies η_i).

Across all configurations tested, the measured SEFD ratios agree with theoretical values at the $\sim 5\text{--}10\%$ level for small baseline groups, and better than a few percent when averaging over many baselines. This confirms that the injected noise and the per-baseline efficiency scaling together reproduce the correct heterogeneous-array sensitivity.

5 PB IMAGE / VOLTAGE PATTERN

CASA uses the complex *voltage pattern* (VP) of each antenna, not the power primary beam (PB) directly. They are related by

$$\text{PB}(\theta) = |\text{VP}(\theta)|^2, \quad (12)$$

so that the VP is the complex field (amplitude illumination pattern) and PB is the corresponding power response.

5.1 Creating correct PBs/VPs for arbitrary dish sizes

Step 1 – Theoretical power PB

Following the ALMA convention (Cortes et al., 2025, 18–19), the power PB is approximated by a Gaussian with

$$\text{FWHM}_{\text{power}} = 1.13 \frac{\lambda}{D}. \quad (13)$$

We denote this target power pattern by $\text{pb_power}(\theta)$.

Step 2 – Convert to voltage pattern

CASA expects a voltage pattern image $\text{VP}(\theta)$ such that $\text{PB}(\theta) = |\text{VP}(\theta)|^2$. Starting from the Gaussian power PB above, we therefore define

$$\text{VP}(\theta) = \sqrt{\text{pb_power}(\theta)}. \quad (14)$$

For a Gaussian this implies

$$\text{FWHM}_{\text{amp}} = \frac{\text{FWHM}_{\text{power}}}{\sqrt{2}}, \quad (15)$$

so that squaring the amplitude pattern recovers the desired power PB with FWHM $1.13 \lambda/D$.

Step 3 – Store VP (not PB) in a .im file

We write out the voltage pattern to an image, e.g.

$$\text{PB_25m.im} \equiv \text{VP}_{25\text{m}}(\theta), \quad \text{PB_12m.im} \equiv \text{VP}_{12\text{m}}(\theta), \quad (16)$$

and register them with CASA via

$$\text{vp.setpbimage(telescope='ALMA', realimage='PB_25m.im', antnames=antlist),} \quad (17)$$

where `antnames` is the list of antennas with $D = 25$ m that should use this PB pattern.

Step 4 – CASA internally squares the VP

During AW-projection gridding in `tclean`, CASA forms

$$\text{PB}_{\text{CASA}}(\theta) = |\text{VP}(\theta)|^2 = \text{pb_power}(\theta), \quad (18)$$

so the primary-beam images (`.pb`) produced by `tclean` for each dish diameter have the expected theoretical HPBW.

Validation – Does this work?

We define two heterogeneous arrays consisting of (i) 2×12 m and 2×7 m antennas (`demo1`), as well as (ii) 10×12 m and 4×25 m antennas (`demo2`). We run the pipeline and investigate the created pb images after `tclean`, and compare the HPBW we measure from them with expected values based on theory. See the following figures for the results.

REFERENCES

- Cortes, P. C., J. Carpenter, S. Kameno, R. Loomis, B. Vila-Vilaro, K. Immer, C. Vlahakis, J. Law, F. Stoehr, K. Saini, and A. Hales (2025). *ALMA Technical Handbook, ALMA Doc. 12.3, ver. 1.5*. ALMA. ISBN: 978-3-923524-66-2. DOI: [10.5281/zenodo.14933753](https://doi.org/10.5281/zenodo.14933753). URL: <https://doi.org/10.5281/zenodo.14933753>.
- Thompson, A. Richard, James M. Moran, and George W. Swenson (1986). *Interferometry and synthesis in radio astronomy*.

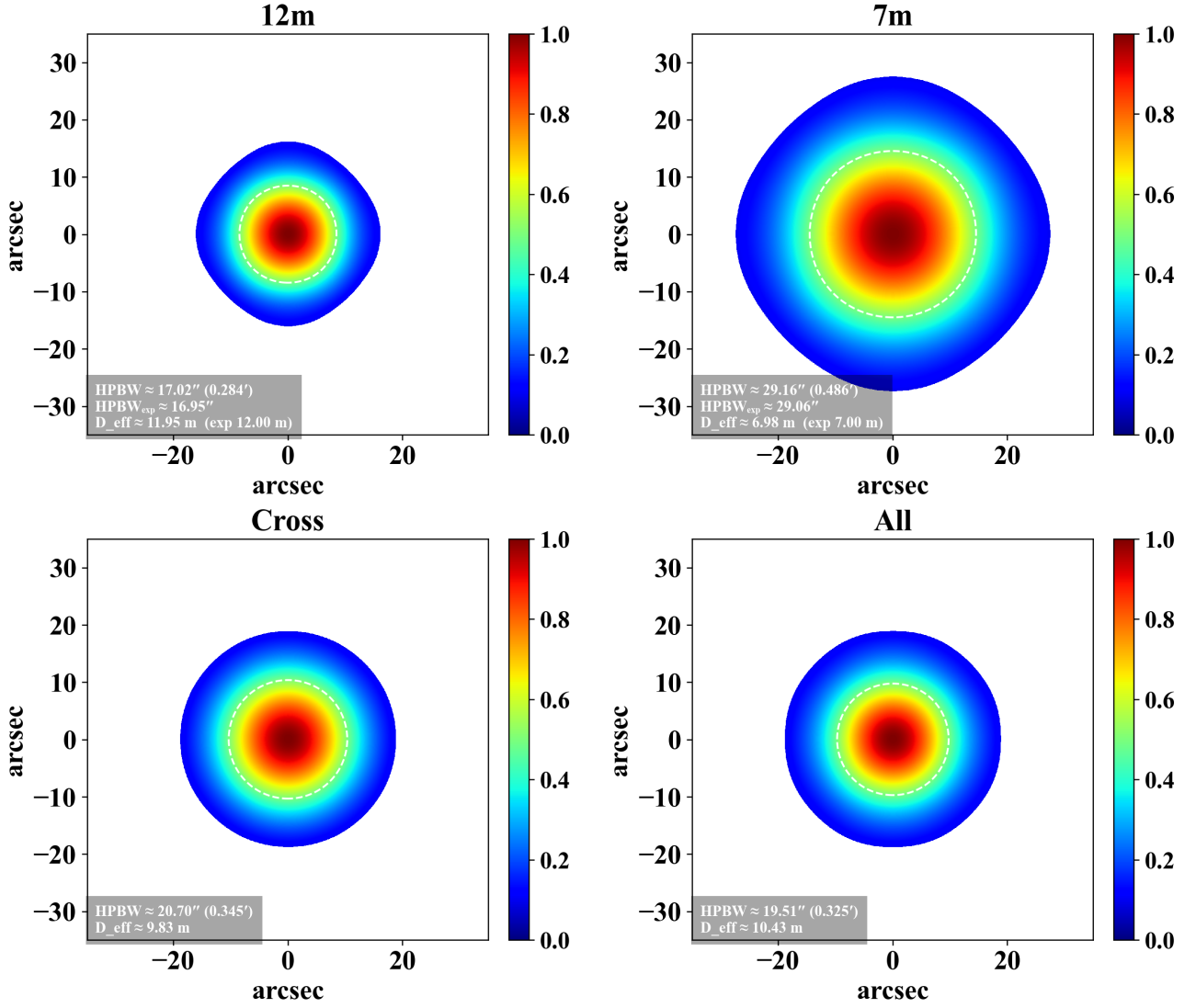


Figure 8: Primary-beam power patterns for the heterogeneous demo1 array (2×12 m and 2×7 m). The maps show the PB derived by `tclean` when supplying per-antenna voltage-pattern images through `vp.setpbimage`. Each panel shows the normalized PB, its measured HPBW, and the theoretical Gaussian expectation $\text{HPBW} \approx 1.13 \lambda/D$ for the corresponding aperture. The “Cross” and “All” beams arise naturally from the heterogeneous array response.

Table 8: Measured primary-beam HPBW compared to the Gaussian $\text{HPBW} \approx 1.13 \lambda/D$ expectation at $\nu = 343.5$ GHz.

PB type	D_{eff} (m)	HPBW_{th} (arcsec)	$\text{HPBW}_{\text{meas}}$ (arcsec)
12m	11.98	16.95	16.98
25m	24.95	8.14	8.15
Cross	19.61	—	10.37
All	19.70	—	10.33

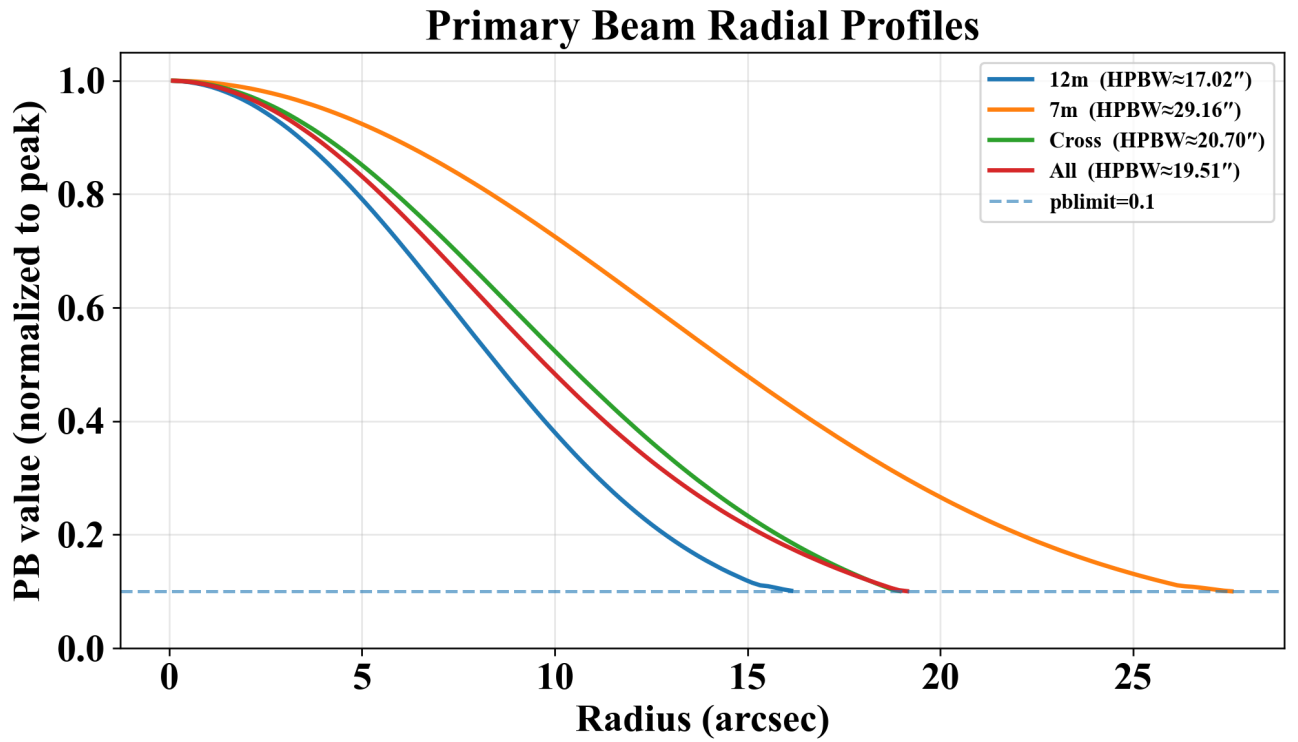


Figure 9: Azimuthally averaged radial profiles of the demo1 primary beams. The curves show the measured PB (normalized to unity) for the 12 m, 7 m, the heterogeneous “Cross” combination, and the overall “All” PB. The `pblimit` used in `tclean` is marked as the striped line. The excellent agreement with the Gaussian $\text{HPBW} \approx 1.13 \lambda/D$ relation confirms that CASA correctly interprets the supplied voltage patterns.

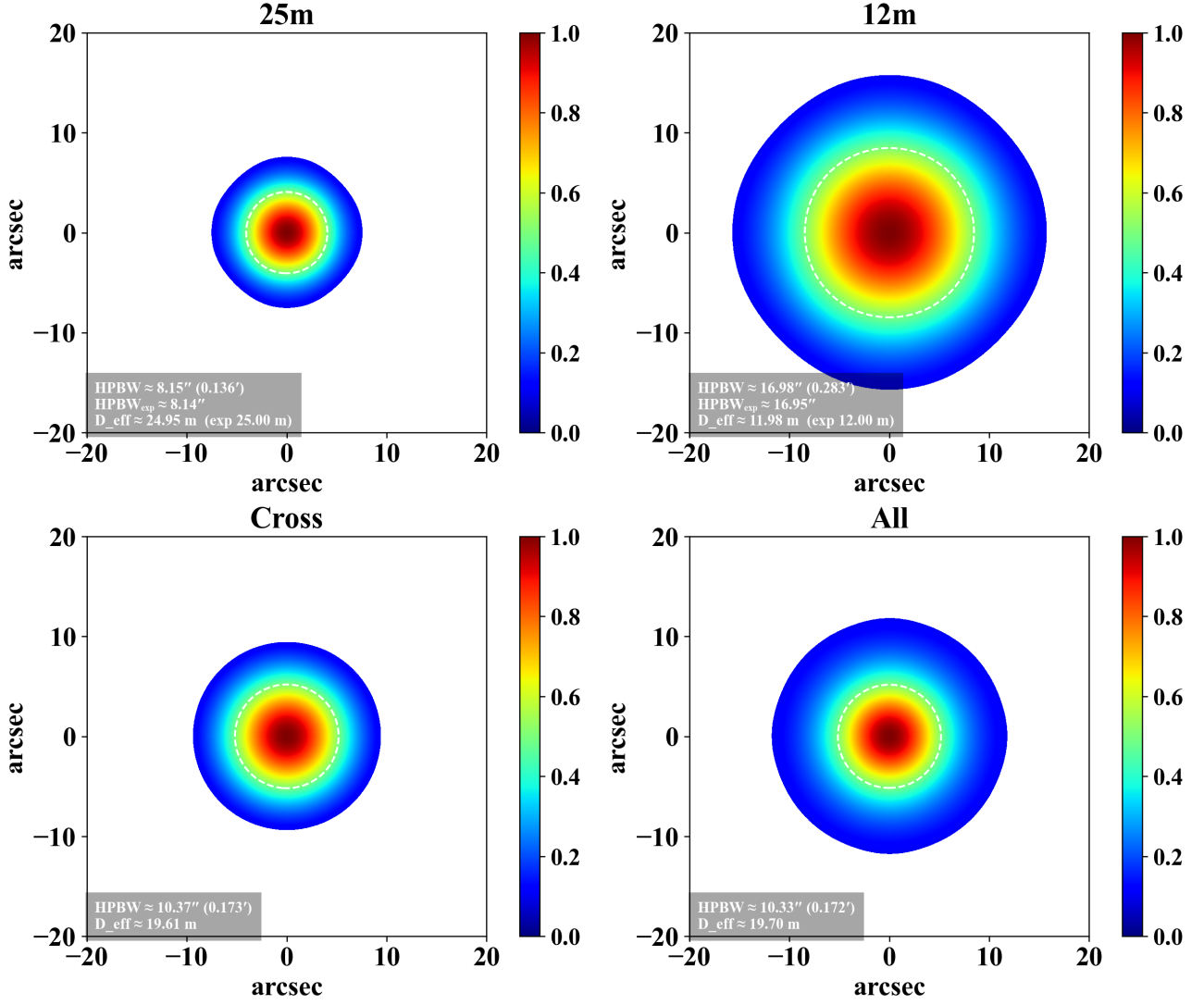


Figure 10: Primary-beam power maps for the demo2 array (10×12 m and 4×25 m). As in demo1, the beams are produced by `tclean` using the per-antenna voltage patterns generated by our PB-construction pipeline. The heterogeneous “Cross” and “All” beams exhibit intermediate widths corresponding to the effective aperture of the mixed array.

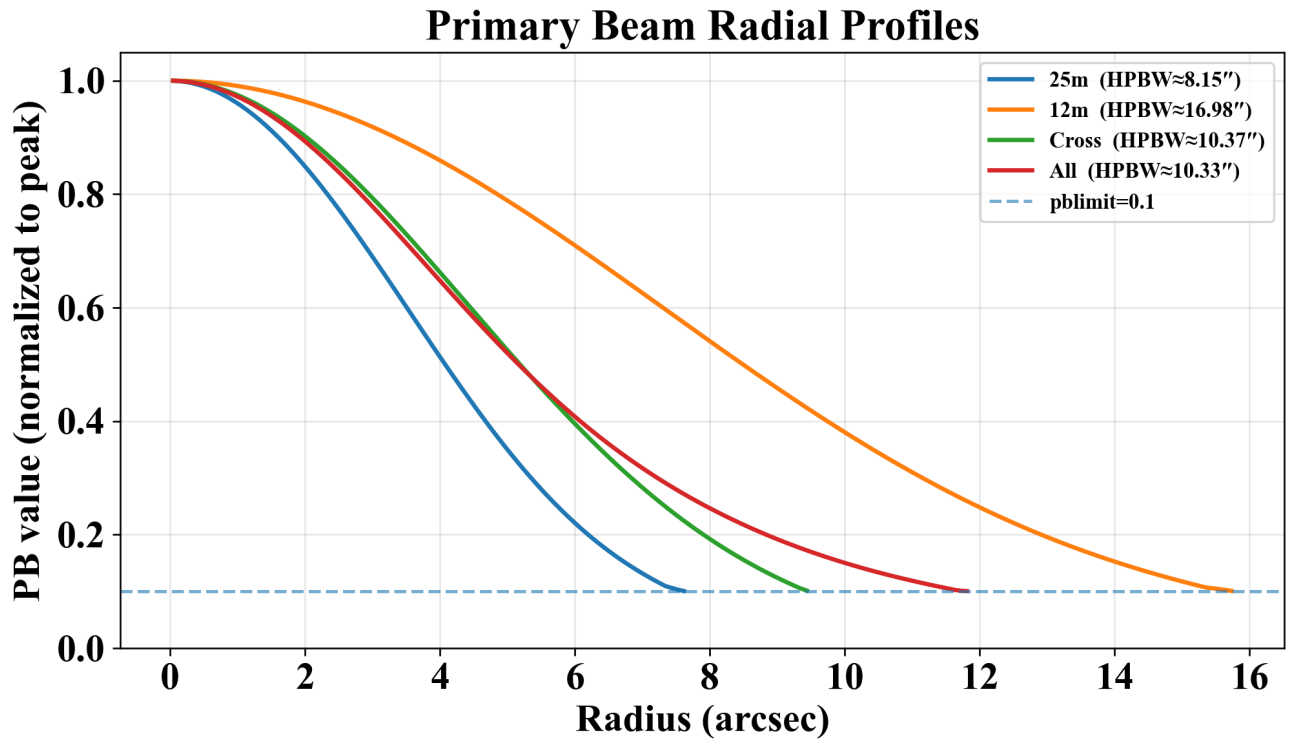


Figure 11: Radial PB profiles for the demo2 configuration. The 12 m and 25 m primary beams follow the Gaussian λ/D expectation, while the heterogeneous “Cross” and “All” PBs fall naturally between them. The dashed horizontal line marks the commonly used primary-beam cutoff (0.1).

EXTRA TEST CFGS + VALIDATION

extra1

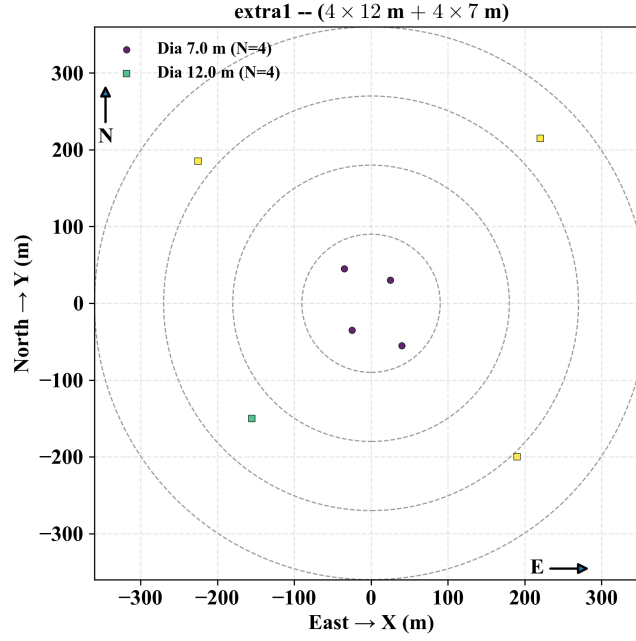


Figure 12: configuration *extra1* – consisting of 4×12 m + 4×7 m antennas.

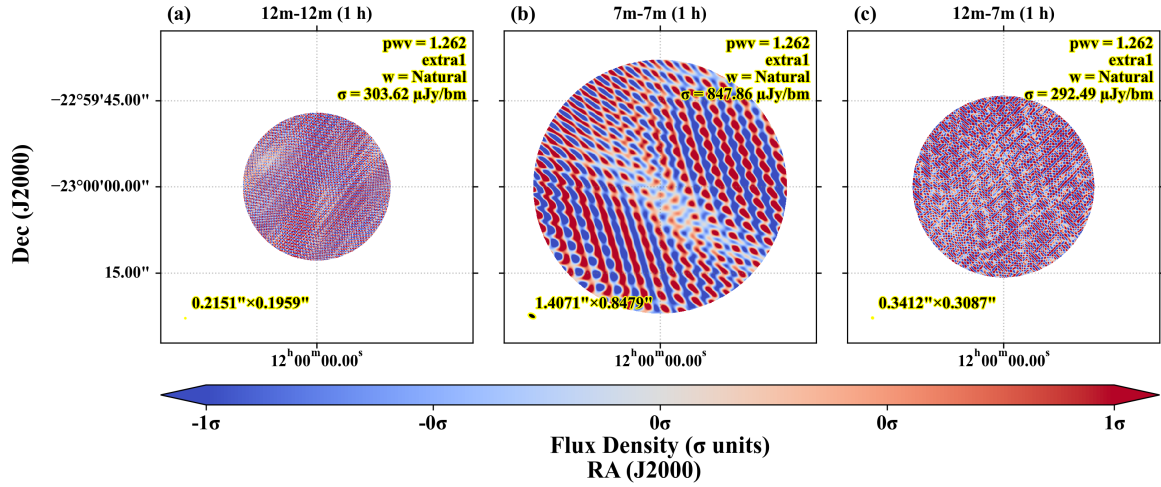


Figure 13: Noise-only images for the heterogeneous array in *extra1* (4×12 m + 4×7 m).

Table 9: Validation metrics for mixed 12 m and 7 m baseline subsets (natural weighting). $\eta_{12} = 0.63$; $\eta_7 = 0.66$.

Baseline	Visibility Statistics				Points	Weights		Image RMS (mJy)		Ratio meas/calc
	VisMean	VisStd				w	σ			
		sim	norm	calc				calc	sim	
(1 h): 12m–12m	0.0255	0.0134	1.000	1.0000	4320	1.0000	1.0000	0.3099	0.3037	0.980
(1 h): 7m–7m	0.0700	0.0368	2.7463	2.9388	4320	0.1158	0.1271	0.8509	0.8479	0.996
(1 h): 12m–7m	0.0423	0.0220	1.6418	1.7143	11520	0.3403	0.3565	0.3141	0.2925	0.931
(1 h): All	—	—	—	—	—	—	—	0.2135	0.2036	0.954

extra2

Now we keep the same 12 m antennas in Fig. 12, but we make all the 7 m antennas 25 m, and re-distribute them:

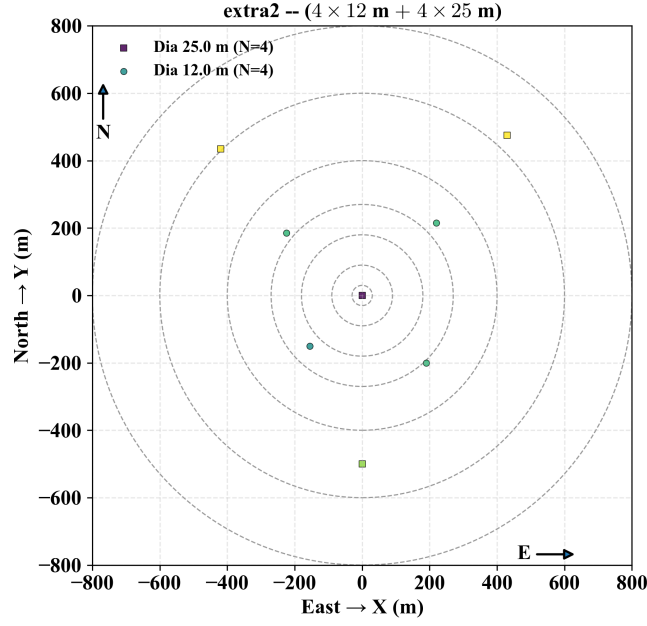


Figure 14: configuration *extra2* – consisting of 4×12 m + 4×25 m antennas.

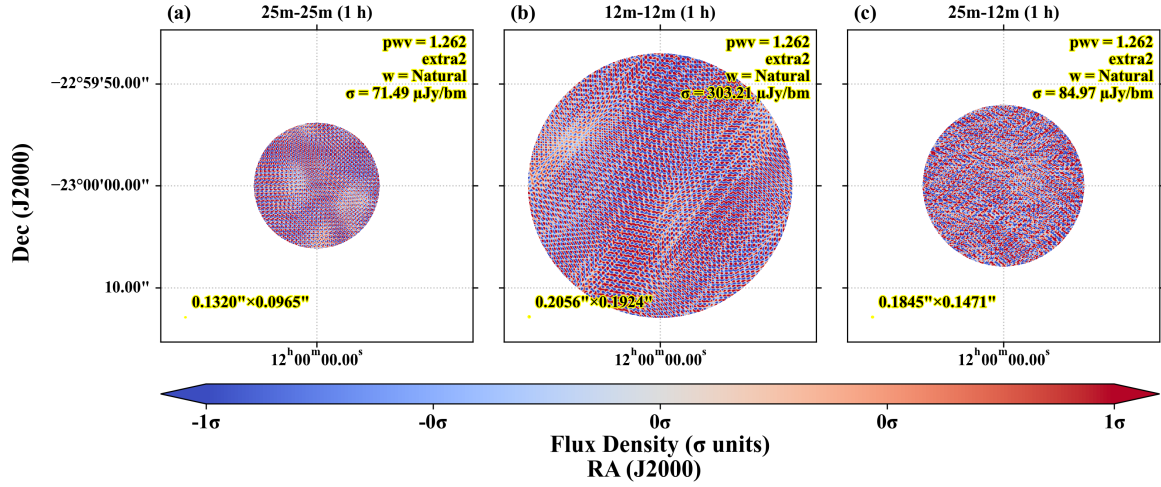


Figure 15: Noise-only images for the heterogeneous array in *extra2* (4×12 m + 4×25 m).

Table 10: Validation metrics for mixed 25 m and 12 m baseline subsets (natural weighting). $\eta_{25} = 0.567$, $\eta_{12} = 0.63$.

Baseline	VisMean	Visibility Statistics			Points	Weights		Image RMS (mJy)		Ratio meas/calc
		sim	VisStd norm	calc		w calc	sim	σ calc	meas	
(1 h): 25m–25m	0.0064	0.0034	1.000	1.0000	4320	1.0000	1.0000	0.0777	0.0715	0.920
(1 h): 12m–12m	0.0255	0.0134	3.9412	4.3403	4320	0.0531	0.0655	0.3099	0.3032	0.979
(1 h): 25m–12m	0.0128	0.0066	1.9412	2.0833	11520	0.2304	0.2560	0.0949	0.0850	0.896
(1 h): All	—	—	—	—	—	—	—	0.0590	0.0548	0.929

extra3

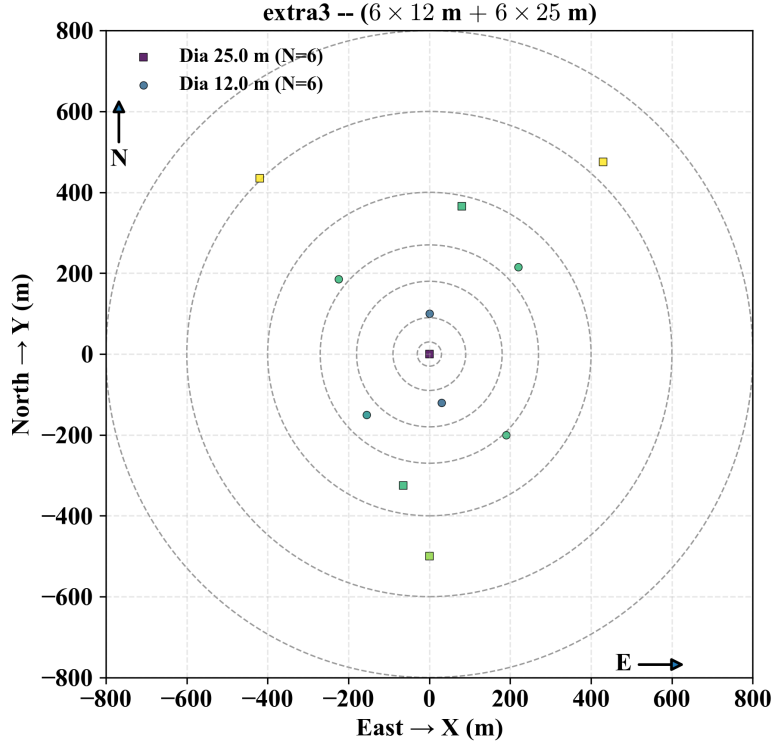


Figure 16: configuration *extra3* – consisting of 6×12 m + 6×25 m antennas.

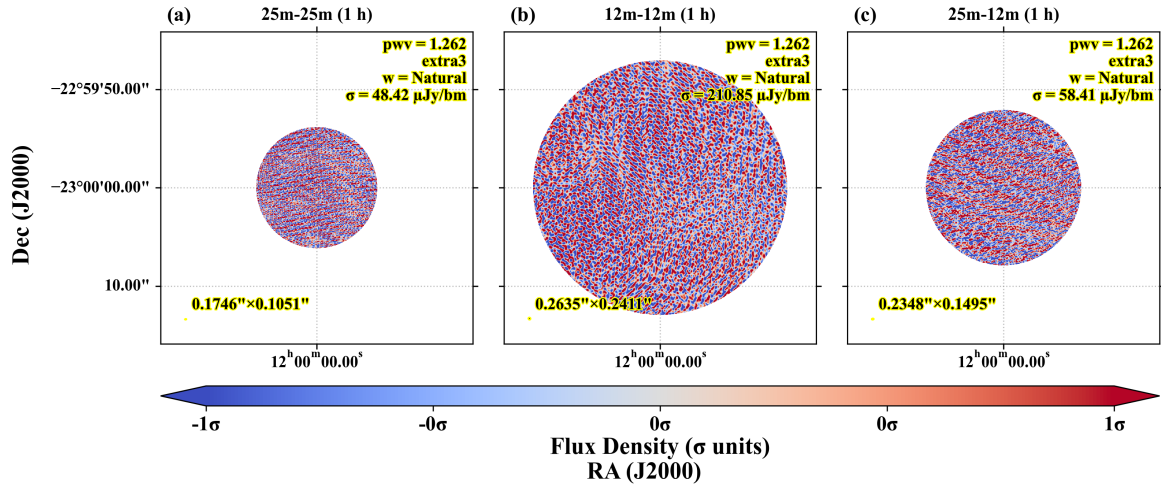


Figure 17: Noise-only images for the heterogeneous array in *extra3* (6×12 m + 6×25 m). $\eta_{25} = 0.567$, $\eta_{12} = 0.63$.

Table 11: Validation metrics for mixed 25 m and 12 m baseline subsets (cross-only; natural weighting, $\eta_{25} = 0.567$, $\eta_{12} = 0.63$).

Baseline	Visibility Statistics				Points	Weights		Image RMS (mJy)		Ratio meas/calc
	VisMean	VisStd		w		σ				
		sim	norm			calc	calc	sim	calc	
(1 h): 25m–25m	0.0064	0.0033	1.000	1.0000	10800	1.0000	1.0000	0.0491	0.0484	0.986
(1 h): 12m–12m	0.0250	0.0132	4.0000	4.3403	10800	0.0531	0.0655	0.1926	0.2109	1.095
(1 h): 25m–12m	0.0127	0.0066	2.0000	2.0833	25920	0.2304	0.2560	0.0630	0.0584	0.927
(1 h): All	—	—	—	—	—	—	—	0.0380	0.0370	0.974

All (extra1, extra2, extra3)

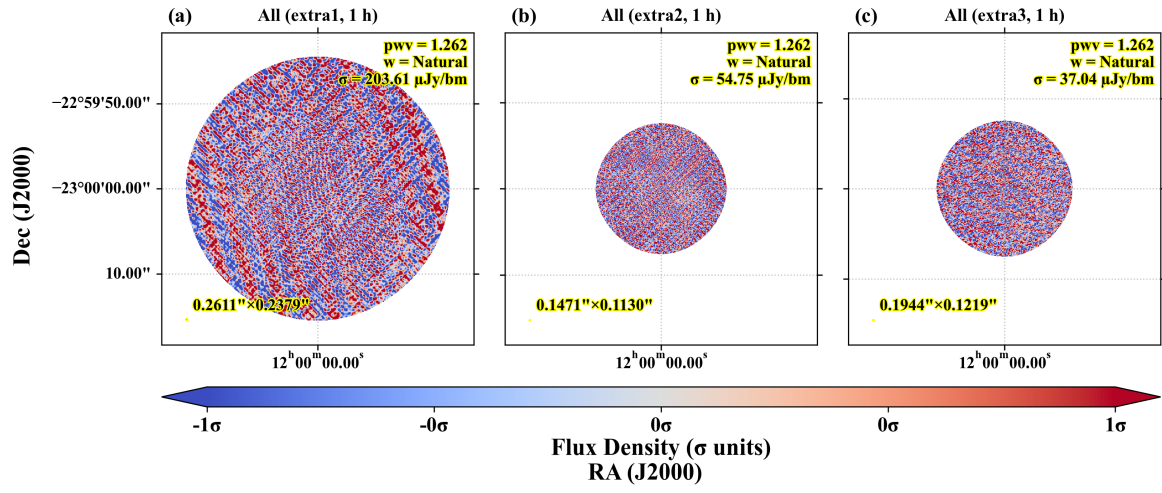


Figure 18: Noise-only images for All the baselines in extra1, extra2, and extra3.